UTM
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Special Topic on AI/ML Tools

Microsoft Azure

Assoc. Prof. Dr. Haza Nuzly Abdull Hamed

AI/ML Tools

Microsoft Azure, Azure Studio

Google TensorFlow, Google AI

Amazon Machine Learning

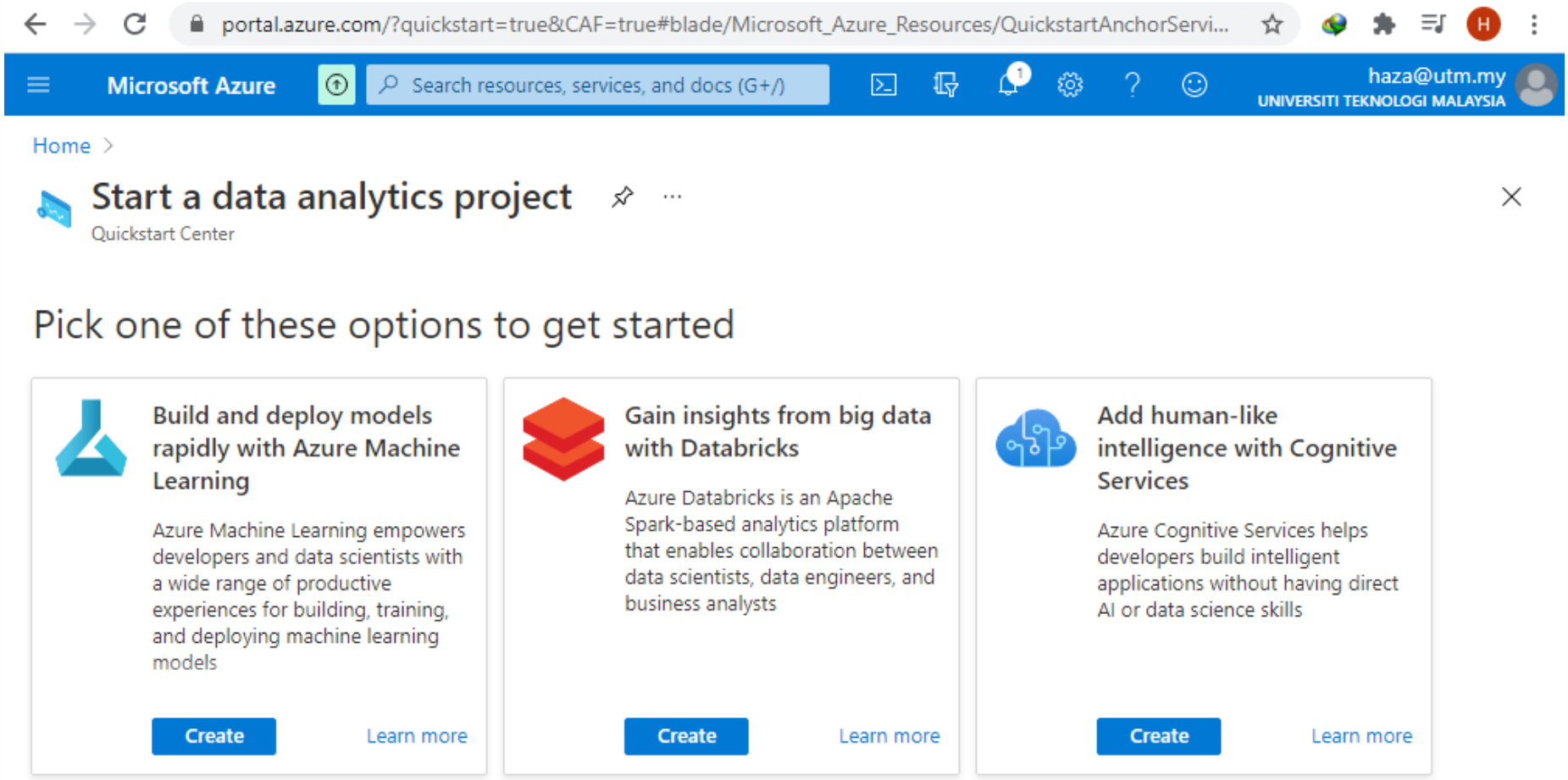
WEKA

Rapid Miner

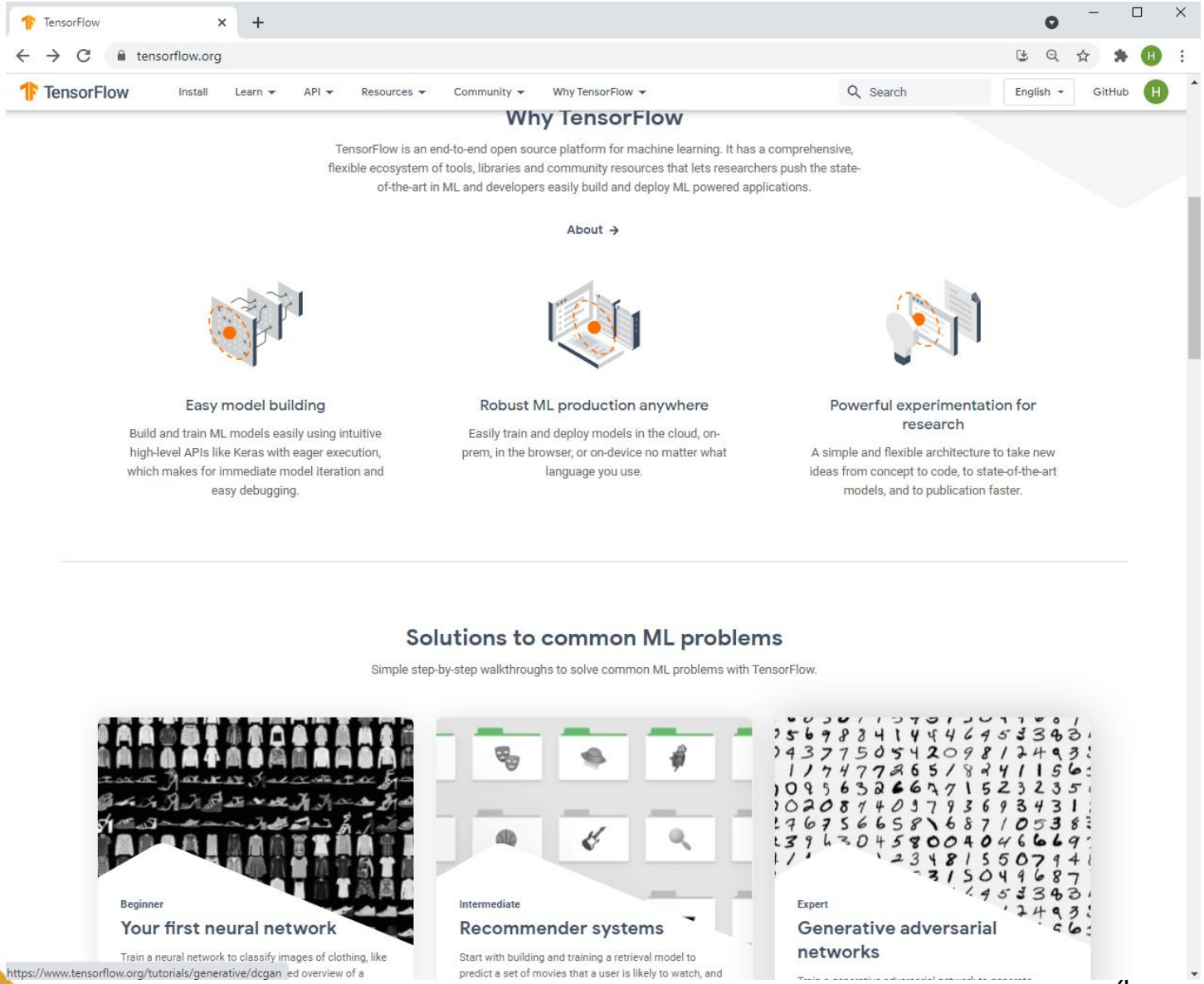
MATLAB

+ Programming in C++ (OpenNN), Python, etc

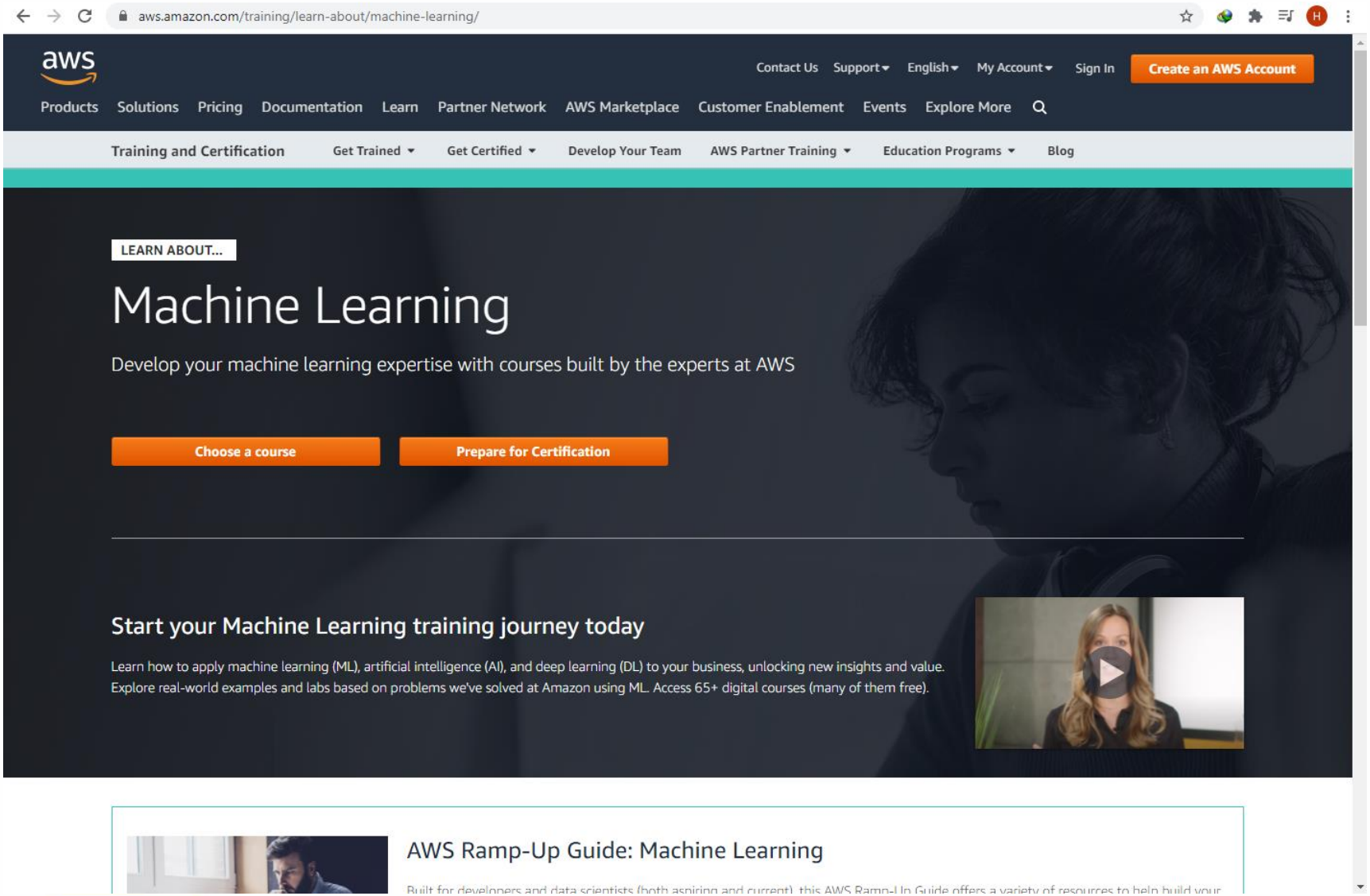
Azure


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Pick one of these options to get started
Build and deploy models rapidly with Azure Machine Learning
Azure Machine Learning empowers developers and data scientists with a wide range of productive experiences for building, training, and deploying machine learning models
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Azure Databricks is an Apache Spark-based analytics platform that enables collaboration between data scientists, data engineers, and business analysts
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Azure Cognitive Services helps developers build intelligent applications without having direct AI or data science skills
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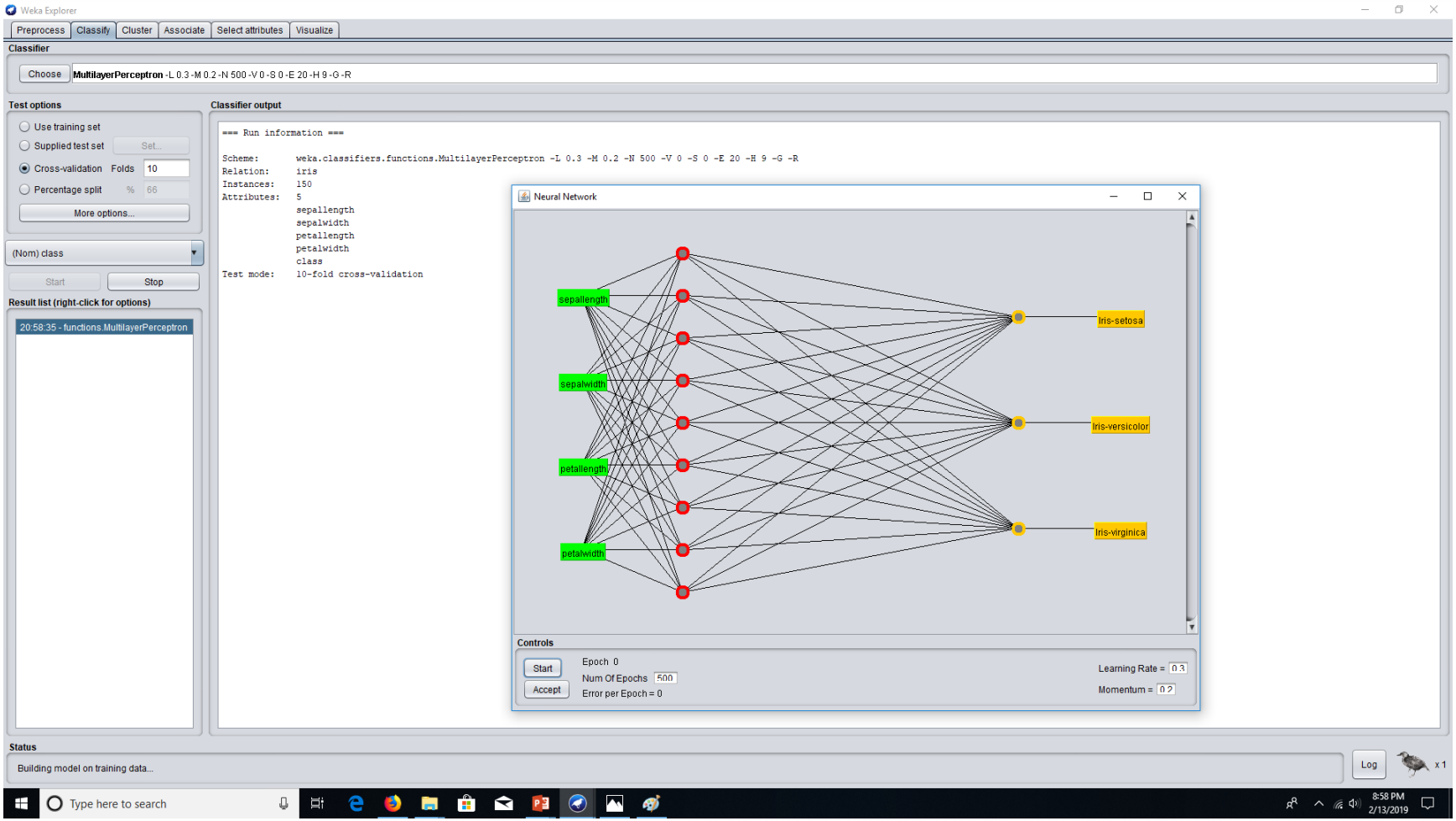
TensorFlow


TensorFlow
tensorflow.org
TensorFlow
Install
Learn
API
Resources
Community
Why TensorFlow
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Why TensorFlow
TensorFlow is an end-to-end open source platform for machine learning. It has a comprehensive, flexible ecosystem of tools, libraries and community resources that lets researchers push the state-of-the-art in ML and developers easily build and deploy ML powered applications.
About
Easy model building
Build and train ML models easily using intuitive high-level APIs like Keras with eager execution, which makes for immediate model iteration and easy debugging.
Robust ML production anywhere
Easily train and deploy models in the cloud, on-prem, in the browser, or on-device no matter what language you use.
Powerful experimentation for research
A simple and flexible architecture to take new ideas from concept to code, to state-of-the-art models, and to publication faster.
Solutions to common ML problems
Simple step-by-step walkthroughs to solve common ML problems with TensorFlow.
Beginner
Your first neural network
Intermediate
Recommender systems
Expert
Generative adversarial networks
https://www.tensorflow.org/tutorials/generative/dcgan

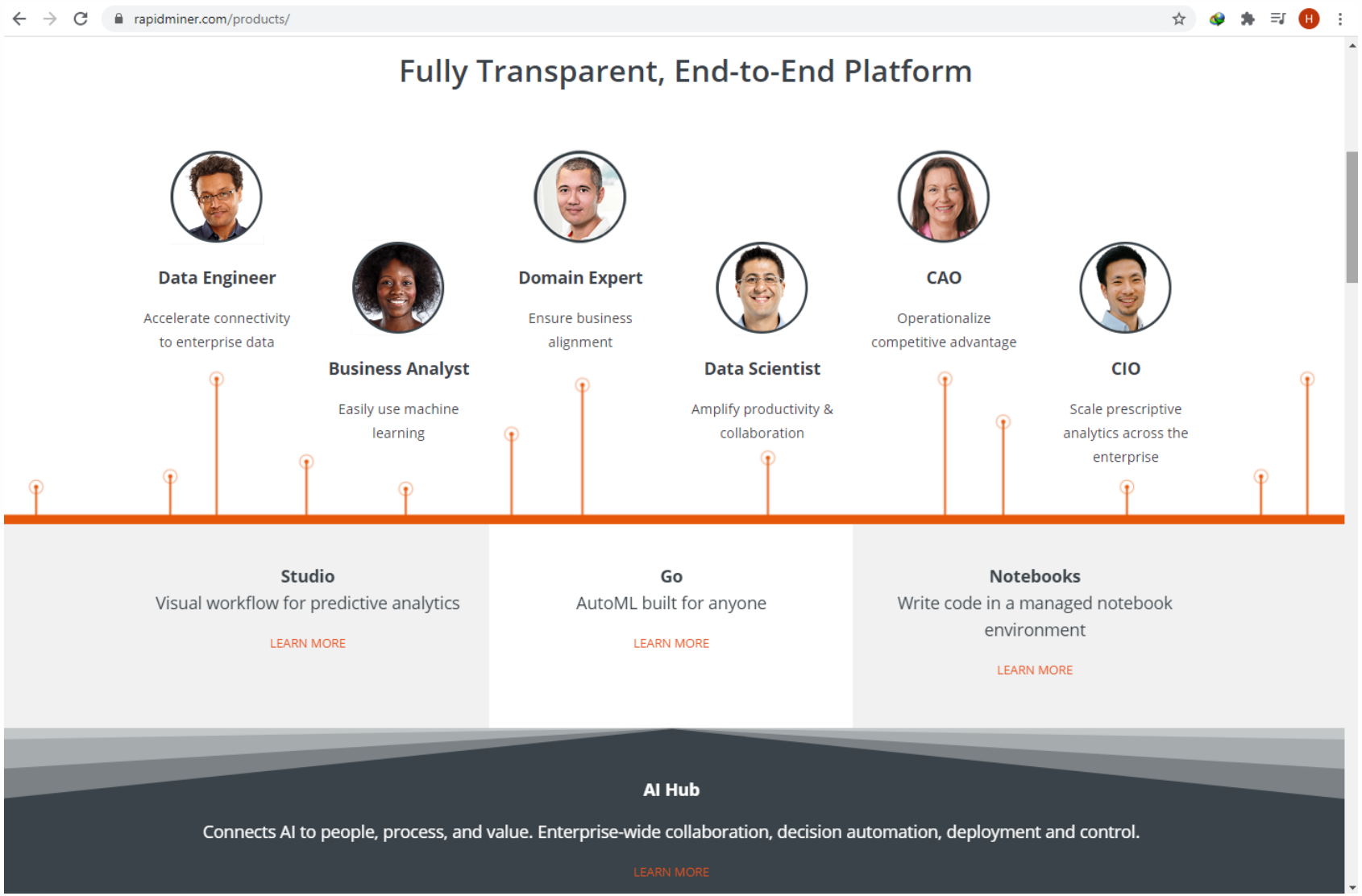
Amazon Web Services



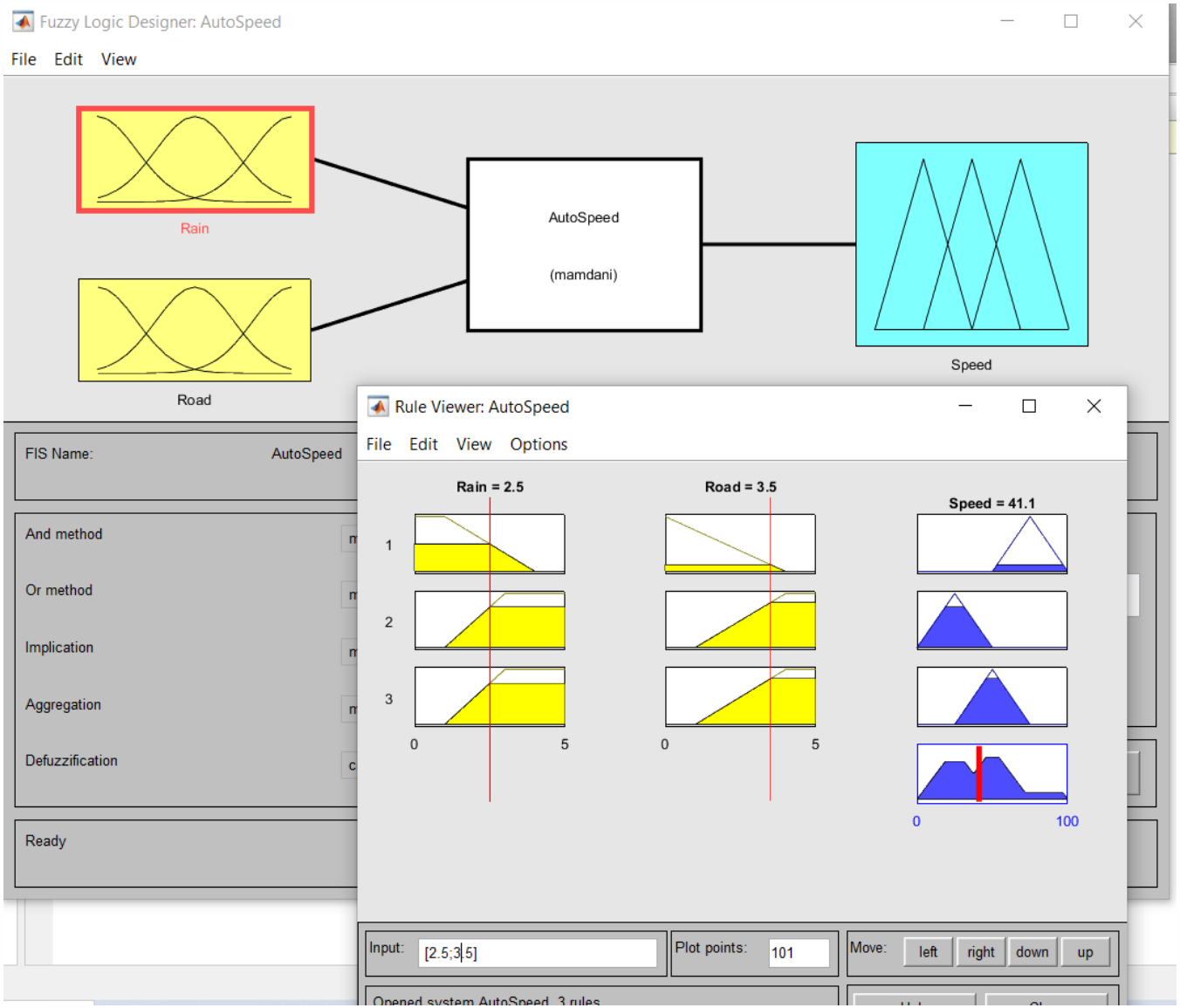
WEKA


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H 9 -G -R
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) class
Start
Stop
Result list (right-click for options)
20:58:35 - functions.MultilayerPerceptron
Classifier output
=== Run information ===
Scheme: weka.classifiers.functions.MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H 9 -G -R
Relation: iris
Instances: 150
Attributes: 5
sepallength
sepalwidth
petallength
petalwidth
class
Test mode: 10-fold cross-validation
Neural Network
sepallength
sepalwidth
petallength
petalwidth
Iris-setosa
Iris-versicolor
Iris-virginica
Controls
Start
Accept
Epoch 0
Num Of Epochs 500
Error per Epoch = 0
Learning Rate = 0.3
Momentum = 0.2
Status
Building model on training data...
Log
x 1
Type here to search
8:58 PM
2/13/2019

RapidMiner

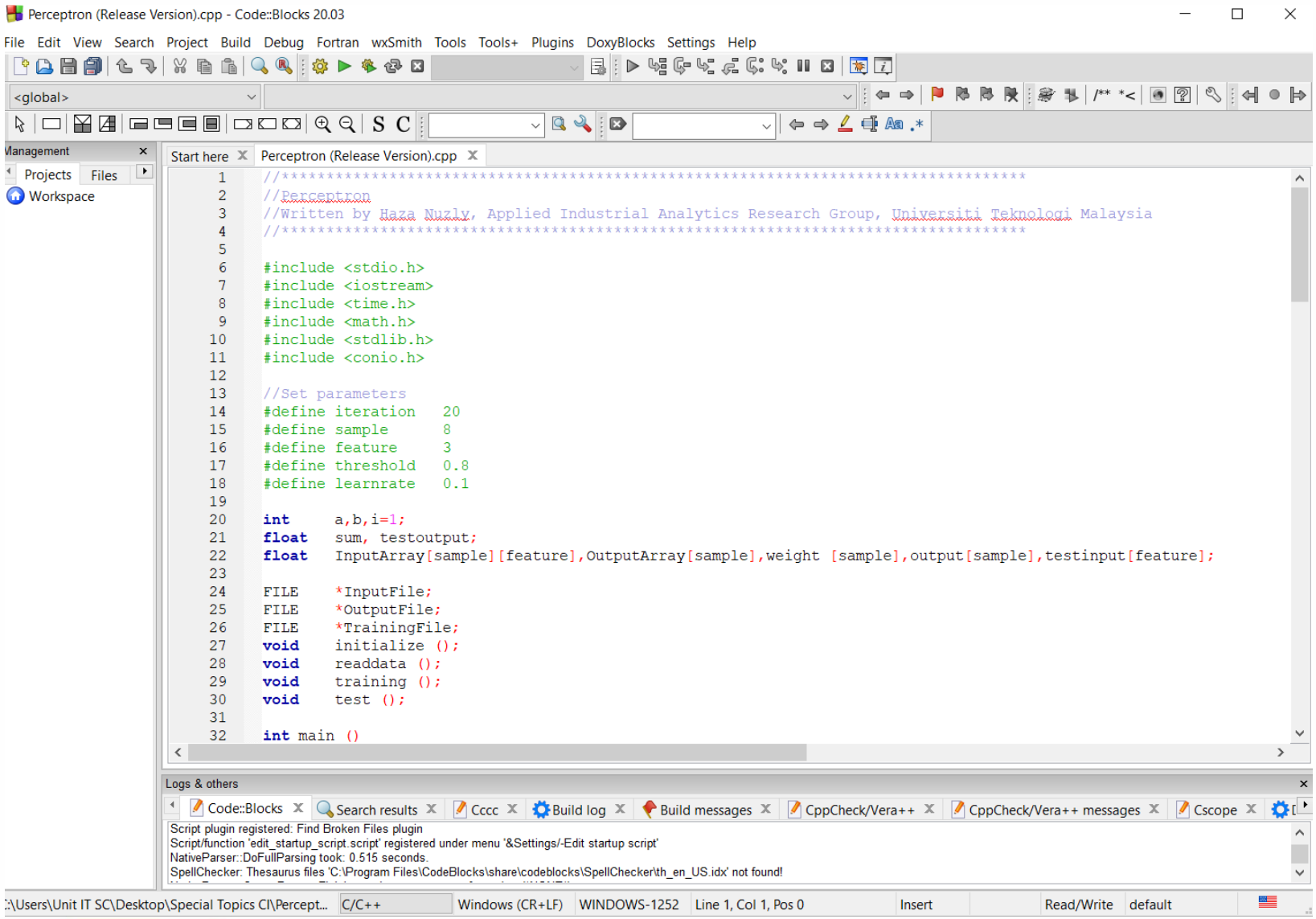

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Business Analyst
Easily use machine learning
Domain Expert
Ensure business alignment
Data Scientist
Amplify productivity & collaboration
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MATLAB



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C/C++



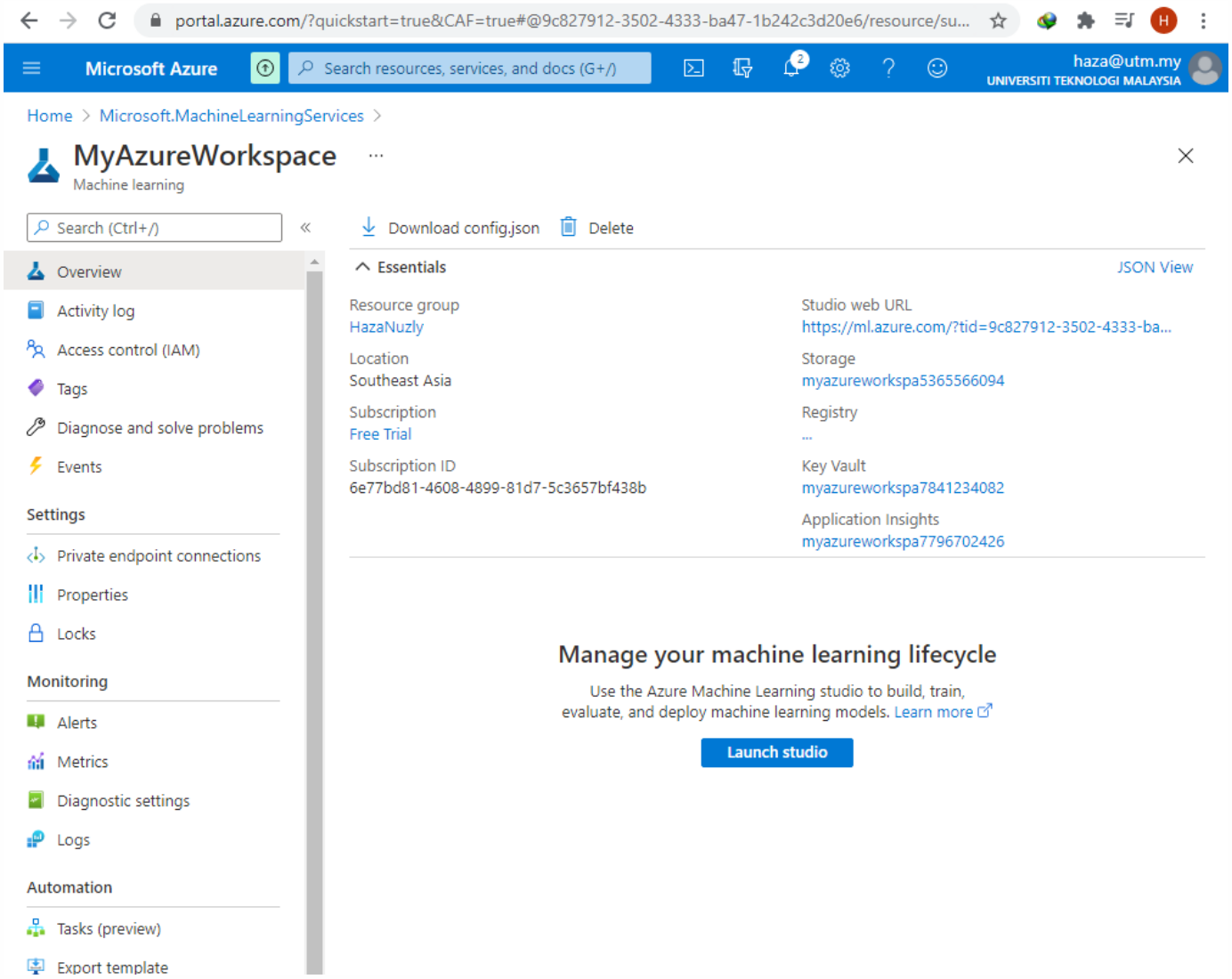
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Microsoft Azure

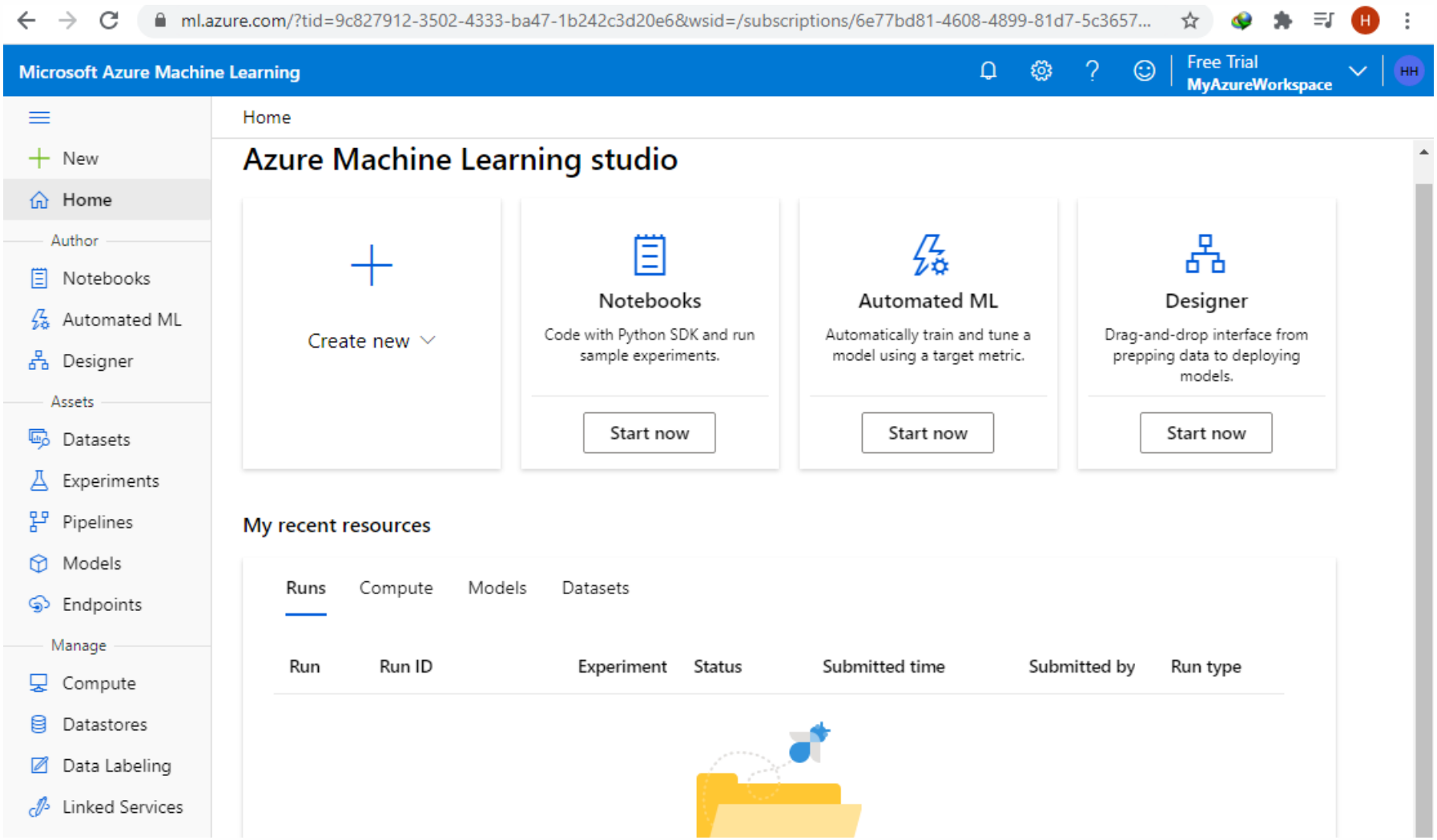
References:

https://docs.microsoft.com/en-us/azure/machine-learning/tutorial-designer-automobile-price-train-score

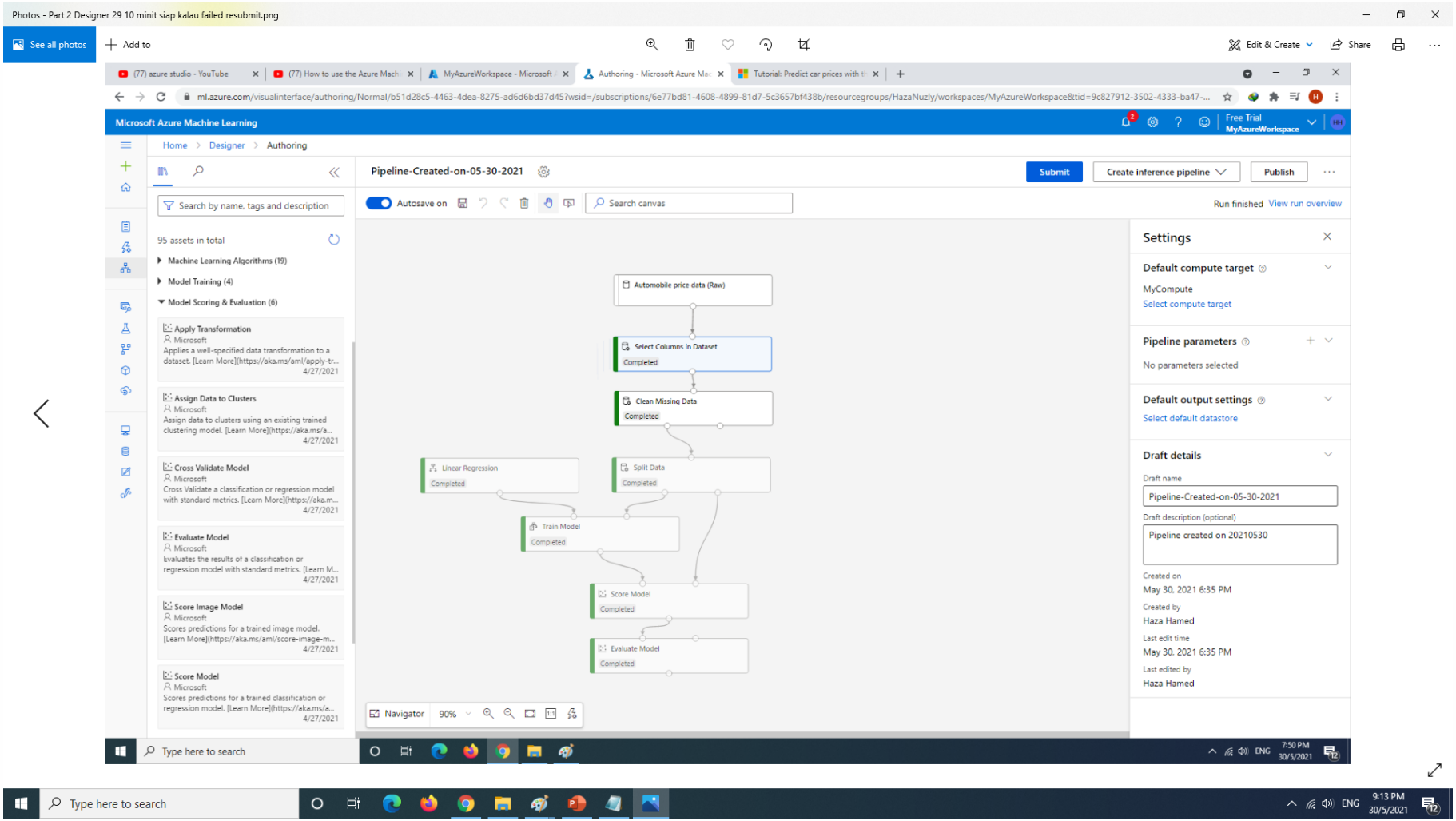
Azure Workspace



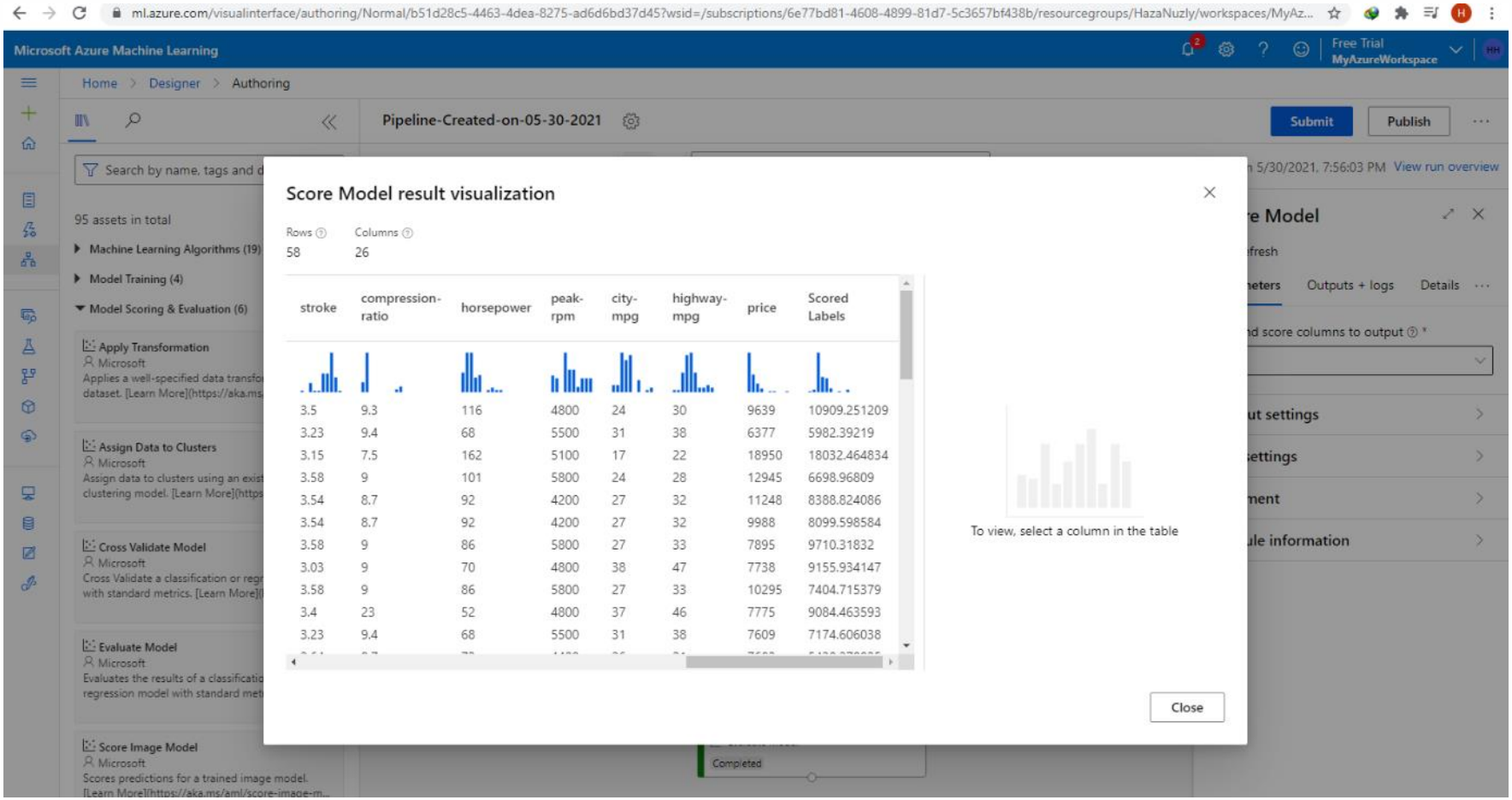
Azure ML Studio



Azure Designer



Azure Designer Result



TensorFlow

Simple tutorial (Run in Google Colab):
https://www.tensorflow.org/tutorials/keras/classification

References:
https://www.youtube.com/watch?v=FiNgII1wRNk

Q & A